

Guidelines and best practices for the scientific use of global iNaturalist data

Corey T. Callaghan^{1,*}, C.J. Campbell^{2,3}, Jane Widness⁴, Benjamin R. Goldstein⁵, Thomas Mesaglio⁶,
Carrie Seltzer⁷, Tony Iwane⁷, Robert P. Guralnick⁸ and Brittany M. Mason^{1,‡}

¹Department of Wildlife Ecology and Conservation, Fort Lauderdale Research and Education Center, Institute of Food and Agricultural Sciences, University of Florida, Davie, FL 33314, United States

²Bat Conservation International, Austin, TX 78746, United States

³Department of Forest and Wildlife Ecology, University of Wisconsin-Madison, Madison, WI 53706, United States

⁴Department of Anthropology, Yale University, New Haven, CT 06511, United States

⁵International Union for Conservation of Nature (IUCN), Washington, DC 20009, United States

⁶Evolution & Ecology Research Centre, School of Biological, Earth and Environmental Sciences, UNSW Sydney, Sydney 2052, Australia

⁷iNaturalist, San Rafael, CA 94915, United States

⁸Florida Museum of Natural History, University of Florida, Gainesville, FL 32611, United States

*Corresponding author. Corey T. Callaghan, Email: c.callaghan@ufl.edu

‡Denotes equal contribution.

Corey T. Callaghan (c.callaghan@ufl.edu) and Brittany M. Mason are affiliated with the Department of Wildlife Ecology and Conservation at the University of Florida, stationed at the Fort Lauderdale Research and Education Center, Institute of Food and Agricultural Sciences, in Davie, Florida, in the United States. C.J. Campbell is affiliated with the Bat Conservation International in Austin, Texas, and the Department of Forest and Wildlife Ecology at the University of Wisconsin-Madison, in Madison, Wisconsin, in the United States. Jane Widness is affiliated with the Department of Anthropology at Yale University, in New Haven, Connecticut, in the United States. Benjamin R. Goldstein is affiliated with the International Union for Conservation of Nature (IUCN), in Washington, District of Columbia, in the United States. Thomas Mesaglio is affiliated with the Evolution and Ecology Research Centre, in the School of Biological, Earth, and Environmental Sciences at the University of New South Wales Sydney, in Sydney, Australia. Carrie Seltzer and Tony Iwane are affiliated with iNaturalist, in San Rafael, California, in the United States. Robert P. Guralnick is affiliated with the Florida Museum of Natural History at the University of Florida, in Gainesville, Florida, in the United States.

Abstract

iNaturalist data are increasingly used in scientific research, yet researchers often lack clear guidance on how to access, handle, and properly attribute these data. Here, we provide a practical guide for the effective use of iNaturalist data in biodiversity research. We begin with an overview of key data concepts, including taxonomy, geoprivacy, location accuracy, and data availability. We then outline methods for acquiring biodiversity observation data, data summaries, and media, and provide example code in R and Python. We discuss key analytical considerations, including observer bias, the presence-only nature of the data, and appropriate modeling approaches. Finally, we highlight best practices for ensuring reproducibility through proper attribution, versioning, and transparent reporting; and we identify emerging research frontiers. This guide aims to support the responsible and effective use of iNaturalist data to further accelerate biodiversity research.

Keywords iNaturalist, citizen science, recommendations, community science, participatory science

Rationale and scope

The majority of contemporary global biodiversity data are now contributed by millions of participants through participatory science (often referred to as citizen or community science) platforms, with at least 80% of recent contributions to the Global Biodiversity Informa-

tion Facility (GBIF) generated through these contributor communities (Callaghan et al. 2021). Among these communities, contributors to iNaturalist collectively generate the broadest geographic and taxonomic coverage of openly available biodiversity observations (Di Cecco et al. 2021), with the volume of contributed observations grow-

Received: 30 March 2026. **Revised:** 4 August 2026. **Accepted:** 14 August 2026

© The Author(s) 2026. Published by Oxford University Press on behalf of the American Institute of Biological Sciences. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com

ing at a mean rate of 50% per year over the last decade. The volume of scientific research leveraging data from iNaturalist is also increasing at rapid rates, with a tenfold increase in publications using iNaturalist data in the last 5 years (Mason et al. 2025). This growth aligns with an increase in policy-usage of iNaturalist data (e.g., environmental impact statements; Callaghan et al. 2025). Despite this rapid uptake, researchers do not always have training in the best practices for understanding and handling iNaturalist data, knowledge of how to efficiently and appropriately access and attribute data, and guidance in how to work with the data most effectively. Because every biodiversity data point originates from the efforts of individual observers and identifiers, responsible scientific use requires guidance on the understanding of the data generation process, as well as the collaborative processes through which those data are generated and maintained.

As with most participatory science datasets, iNaturalist data are uploaded by many different observers from different parts of the world. Attributes inherent to participatory science (e.g., variation in observation effort, identification accuracy, and metadata completeness) can affect and inform the resulting utility of the data. This includes variation in the spatial and temporal accuracy of observations (Feldman et al. 2021), the quality and completeness of image metadata (Pernat et al. 2024a), and the spatial coverage of observations across landscapes (Geurts et al. 2023a). Because the data are used across disciplines—ranging from ecology to computer science to education—there are heterogeneous and diverse uses of, and caveats to, the data depending on the research questions. There is therefore a need to summarize the current best practices and guidelines for accessing the necessary data, understanding its structure, and using the data in practice.

We provide the first comprehensive methodological guide for effectively using global iNaturalist data in scientific research (figure 1). This effort is inspired by similar guides to using eBird data in research (Johnston et al. 2021, Strimas-Mackey et al. 2023). We systematically explain how iNaturalist observations are generated, curated, structured, accessed, and analyzed, with an emphasis on the implications of these processes for biodiversity research. Specifically, we provide an overview of iNaturalist data and foundational concepts (box 1), describe the critical data components and metadata required to interpret observations, outline the major pathways for accessing data (GBIF, the Export Tool, and the API), discuss analytical considerations and best practices for working with this complex, dynamic dataset, and provide guidance on attribution and reproducibility. We additionally provide code examples in R and Python for some of the above, as well as provide perspectives on the future of iNaturalist data usage. Although our focus is on biodiversity research, we believe these guidelines will be pertinent to many different research disciplines (e.g., social sciences, computer science) and anticipate that these guidelines will facilitate the further use of iNaturalist data for scientific research and policy implementation.

Box 1: Foundational concepts for using iNaturalist data.

iNaturalist is a global platform where users can upload photos or audio recordings to the iNaturalist platform through a smartphone or the website as evidence of an organism at a certain location and time. As an observation is uploaded, iNaturalist's computer vision model is available to suggest species-level identifications based on the submitted media and geographic context. The observing user can then

review these candidate suggestions as they develop an initial identification. The community then helps identify the observation to the finest possible taxonomic resolution (Callaghan et al. 2022, Campbell et al. 2023). Consequently, each observation represents the product of an ongoing collaborative process involving observers, identifiers, curators, and other contributors. Although most identifications are made by experienced users (Campbell et al. 2023), there are no requirements to become an identifier, and no extra weight is given to experts.

A key piece of information in iNaturalist data is the “quality grade” of a given record: “Research Grade,” “Needs ID,” or “Casual.” Broadly, Research Grade status is attained when an observation meets the site's metadata quality criteria, and has two or more suggested identifications, more than two-thirds of which agree at a species level or a taxonomic level finer than family when further progress in identification is deemed not possible (i.e., 2/2, 3/3, 3/4, etc.). Observations classified as “Needs ID” meet the metadata requirements for Research Grade but have not yet reached community consensus on their taxonomic identification. Because community identifications accumulate over time, recently uploaded observations are less likely to have reached identification consensus than older observations. Researchers using very recent data should therefore recognize that some observations may still be awaiting community review. Note that Research Grade does not guarantee a correct identification, and researchers should take care to verify observations themselves as appropriate. By contrast, “Casual” observations are those that lack required information (e.g., photo or audio evidence, location, or date), are cultivated, captive, or otherwise not “wild,” cannot be identified to a finer taxonomic level than family, and are marked as such through community annotation. See glossary for more details on these key “observation quality grades.”

The iNaturalist infrastructure provides mechanisms for organizing these different types of observations into structured collections. Most notably, users can create “Projects” (see Glossary), which other users may join or contribute to. Projects allow observations to be grouped, curated, and enriched around a shared research question, geographic region, taxonomic focus, or community initiative. Projects allow for observations to be systematically organized, filtered, and augmented with additional metadata (e.g., through observation fields), enabling coordinated data collection and curation within an otherwise open and dynamic dataset. Observations can be included in an unlimited number of projects.

Another important component of the data submitted to iNaturalist is the licensing. Three types of content can be licensed separately on iNaturalist: (1) observation data (including metadata such as date, geographic location, species identification, etc.), (2) images, and (3) audio recordings (iNatHelp 2024). By default, when creating a new account, all three of these groups are set to CC BY-NC (Creative Commons: Attribution-NonCommercial), meaning that the content can be used with attribution for non-commercial purposes. Other licensing options that users can choose are CC0 (No Copyright), CC BY (Attribution), CC BY-NC-SA (Attribution-NonCommercial-ShareAlike), CC BY-NC-ND (Attribution-NonCommercial-NoDerivs), CC BY-ND (Attribution-NoDerivs), CC BY-SA (Attribution-ShareAlike), and no license (all rights reserved). Licenses are set independently for observations, images, and audio recordings, and can differ within a single observation (figure 2). These licensing settings govern whether and to what extent researchers may legally reuse iNaturalist content. As of July 2026, approximately 23% of iNaturalist observations are not available under an open observation license and therefore can-

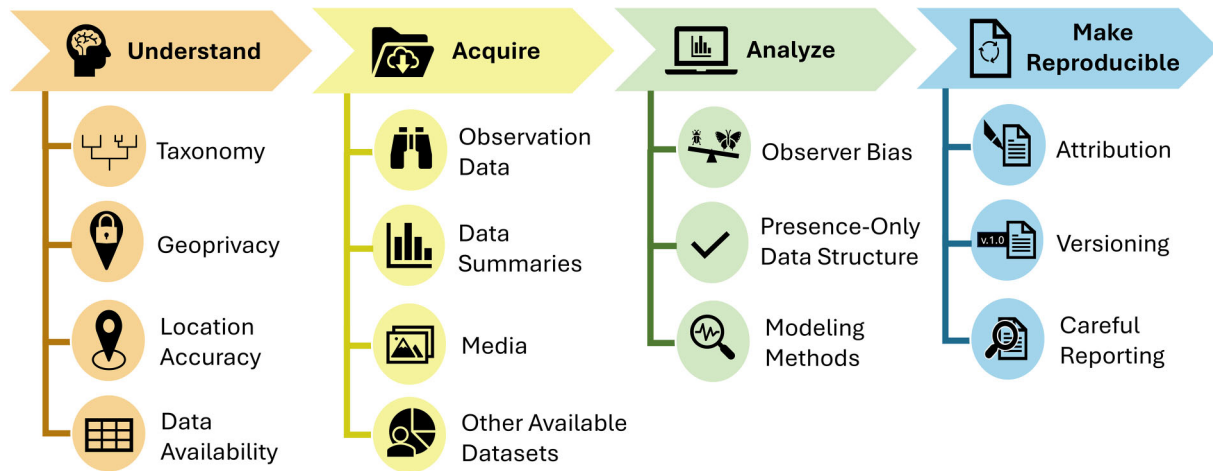


Figure 1. Using iNaturalist data effectively in scientific research requires understanding at multiple levels how to interpret the data, acquire the data, analyze the data, and ensure reproducibility.

not be redistributed through GBIF, highlighting how licensing choices substantially influence the availability of observations for scientific research. If no license is set, then the data are considered “all rights reserved.” iNaturalist generates a weekly export of observations eligible for inclusion in GBIF (Heberling et al. 2021) through which researchers can access biodiversity data. Exported observations include only Research Grade observations that are licensed as CC0, CC BY, or CC BY-NC (default option).

A final important concept to understand about iNaturalist data is that it is dynamic. Identifications, taxonomic definitions, metadata, and even licensing settings can change over time as community members contribute new information or users update their content. As a result, datasets retrieved for scientific research on different dates may differ, and reproducible workflows should explicitly document download dates, filters, and observation identifiers.

Critical data components and metadata for working with iNaturalist data

This section details critical data components and metadata from iNaturalist, including geoprivacy settings, obscuration, data accuracy, and the types of data available on the platform. For components of taxonomic information, see box 2.

Geoprivacy, Obscuration, and Data Accuracy

iNaturalist observations vary in their spatial precision due to geoprivacy settings, coordinate obscuration, user-defined positional uncertainty, and differences in GPS precision, with measures of coordinate uncertainty available through both GBIF and the iNaturalist API or the iNaturalist Export Tool. By default, observations uploaded to iNaturalist have their geoprivacy set to “open,” meaning that the user permits the data to be publicly shared, and other users can freely see the coordinates and date/time of these records. However, some iNaturalist observations have their geoprivacy changed to “obscured,” which can occur in two different ways. First, users may choose to manually obscure any of their records (referred to as “geoprivacy”) at any time for privacy or other personal reasons.

Second, some observations are automatically obscured by iNaturalist (referred to as “taxon geoprivacy”; see Supplemental Text S1 for more information).

When an observation is obscured, iNaturalist hides the original latitude and longitude, replaces the publicly displayed coordinates with a randomly assigned location within a $0.2^\circ \times 0.2^\circ$ grid cell, and sets the public coordinate uncertainty (positional accuracy) to the length of that grid cell’s diagonal (~ 31 km at the equator; decreasing in size toward the poles; Supplemental Text S1–S2). In GBIF, obscured observations are not explicitly flagged; instead, the obscured coordinates and their associated coordinate uncertainty are reported. Information on whether an observation has been obscured is only available from iNaturalist through the geoprivacy metadata field, along with the obscured coordinates and coordinate uncertainty. “Private” observations, a third type of geoprivacy, share no geographic information publicly, and these observations are not shared with GBIF. Users may share obscured or private coordinates by trusting a project or individual user (iNatHelp 2023). Network nodes (i.e., regional or national partners that operate localized versions of the iNaturalist platform; iNaturalist 2026a) also get access to true coordinates of observations made by users who have chosen to affiliate with them. If granted access to these hidden coordinates, take special care to avoid sharing this sensitive data in publications or data releases.

Users may also upload observations with intentionally coarse positional accuracy for the coordinates (Supplemental Text S1). This may occur because users do not remember the precise location at which they made the observation, or as an alternative method to obscure the location.

Data available on iNaturalist

The most commonly used data from iNaturalist are date and time of an observation, consensus taxonomic identification (hereafter Community Taxon), and geographic location (Mason et al. 2025). Spatiotemporal data from iNaturalist are used by researchers for a myriad of reasons, including to describe species ranges and distributions (Cull 2022, Wolfe et al. 2025), assess biodiversity trends (Beninde et al. 2023), compile species lists (Wenk et al. 2024), and more.

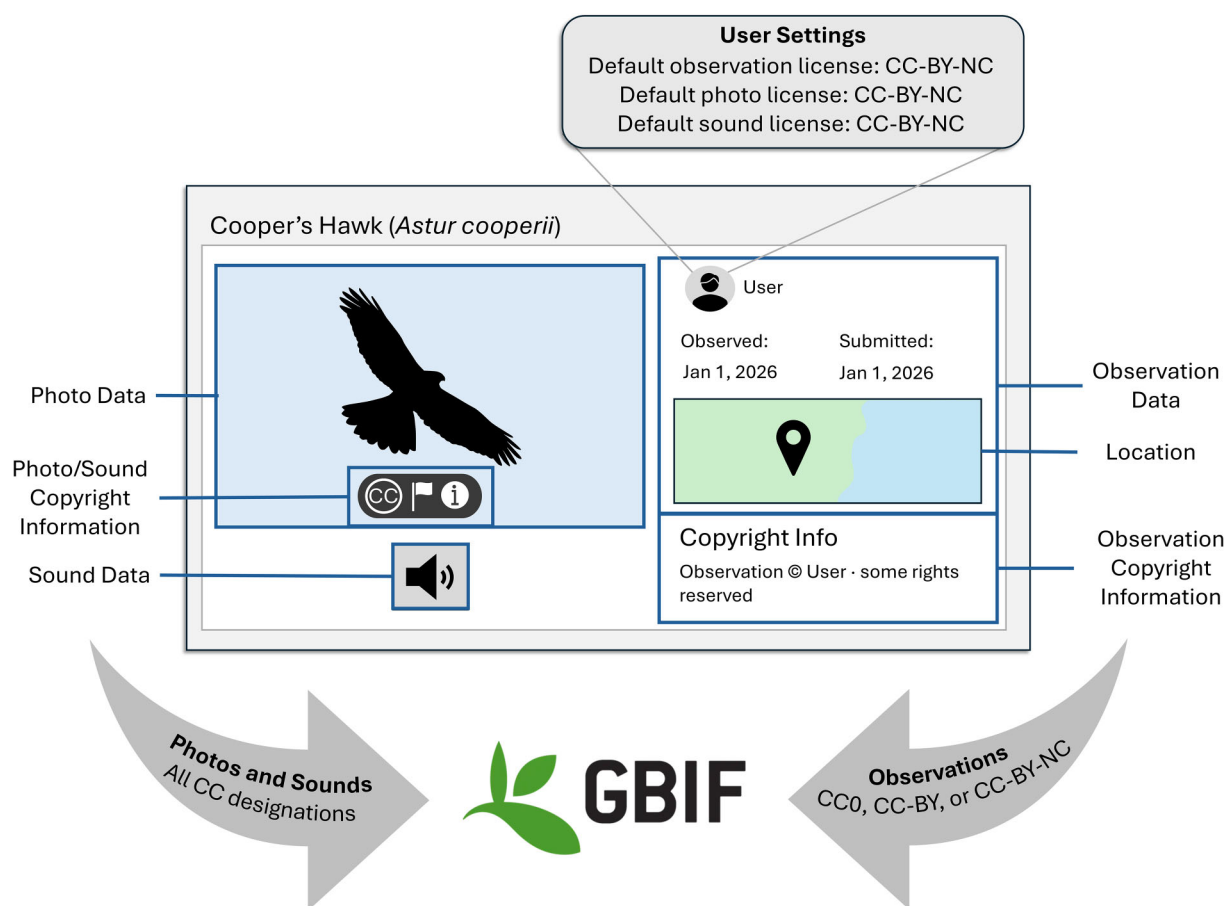


Figure 2. Schematic of an iNaturalist observation from the iNaturalist website illustrating the three independently copyrighted data sources associated with iNaturalist records: photo data, sound data, and observation data. The diagram also shows where copyright and licensing information can be found for each data type. Photos and sounds with any Creative Commons (CC) designation are shared with GBIF for scientific use, whereas observations must be designated as CC0, CC-BY, or CC-BY-NC (the default) to be shared with GBIF. Researchers are responsible for ensuring that data are used in accordance with the applicable licenses.

Box 2: iNaturalist has a dynamic taxonomic backbone.

iNaturalist's taxonomy is highly dynamic, with changes—including the addition, synonymization, and splitting of taxa—implemented daily by site curators to reflect changes in both primary (the scientific literature) and secondary (taxonomic database) sources. Broadly, curators attempt to align iNaturalist taxonomy to global taxonomic authorities for taxa where these exist, such as Plants of the World Online (Royal Botanic Gardens Kew 2026) for vascular plants or The World Spider Catalog (Natural History Museum Bern 2026) for spiders; a [list of these external authorities](#) is available for curators on the site (see [How Taxon Changes Work](#) for complete details on how this affects iNaturalist data). Although following designated external authorities as closely as possible is strongly encouraged, curators can “deviate” from these authorities where appropriate to follow differing regional treatments. Regional taxonomic authorities and the primary literature are referenced for taxa where global taxonomic authorities do not exist, such as fungi or many terrestrial invertebrate groups.

iNaturalist does not maintain or publish historical versions of its taxonomic tree, so if it is to be referenced alongside an iNaturalist data download, both should be obtained at the same time. The latest version of the tree can be downloaded through

the [iNaturalist Taxonomy DarwinCore Archive](#), which is generated monthly. It is important to note that, because iNaturalist is an observational database rather than a nomenclatural database, it is not intended to represent an exhaustive source of all names for all taxa. The iNaturalist taxonomic tree differs from the one used by GBIF. When iNaturalist shares data to GBIF, GBIF uses its own [taxon match service](#) to align taxa with its own taxonomy. Until June 2026, this was [GBIF's taxonomic backbone](#), but it now matches to both the legacy backbone and also to the [Catalogue of Life XR \(COL XR\)](#), the new taxonomic backbone for GBIF (Bánki et al. 2026). When name mismatches occur, GBIF may use fuzzy matching or match to a coarser taxon. The legacy GBIF Backbone Taxonomy was last updated in 2023, and is retained for backward compatibility. The new COL XR Backbone is updated monthly. Both the legacy backbone and the COL XR backbone can be downloaded from <https://www.checklistbank.org/>.

Species lists, summarized species count data, and range maps can be a useful alternative to downloading a full dataset when fine-scale detail is not required. These include summarized species counts for specified parameters (e.g., place, taxon, user) and summarized counts by time (e.g., week of the year, month, year). Furthermore, iNaturalist provides over 100,000 range maps that have been modeled from iNat-

urologist data (see more [here](#), including information on how the maps were made; [Supplemental Text S12](#); iNaturalist 2026b).

Media submitted to iNaturalist are also a rich data source that is increasingly used by researchers (Mason et al. 2025). To extract useful data from images and audio files, such as behavior or morphology, researchers may manually review media and record relevant data, or they may use advanced computational techniques such as supervised classification or computer vision models (Perez-Udell et al. 2023). Audio data are less commonly used in the literature, but examples include studying anuran breeding phenology (Forti et al. 2022) and testing the ability of convolutional neural networks to accurately identify species based on their calls (Wu et al. 2022). Additional data types, such as identification history and user activity trends, are available for special use cases (see [Supplemental Text S1](#) for examples).

Additional data fields that can be added to iNaturalist observations include annotations, observation fields, and tags (see [table 1](#)), which can be accessed through the iNaturalist Export Tool or API but are not available on GBIF. These fields are optional and can be contributed by the original observer or, in the case of annotations and observation fields, by the broader community—unless the observer has restricted who can add fields to their observations. Annotations are a limited set of fields with prescribed values that are decided by the iNaturalist administrators, with options including, but not limited to, “alive or dead,” “evidence of presence,” “life stage,” “sex,” “flowers and fruits,” and “leaves.” Some annotations are only available for certain taxa; for example, the flowers and fruits and leaves fields are only available for plants where those attributes are applicable. Although most records are not annotated, the number of annotated records is increasing each year and can provide valuable insights ([figure 3](#)). Annotations may be useful for studying phenology (e.g., timing of flowering) or providing human labels to train machine learning models (Dinnage et al. 2025), studying differing life stages (e.g., timing of hatchling emergence), or filtering the data (e.g., removing dead animal observations prior to examining phenology). Additional data are available from observation fields, which are often associated with specific projects that seek to gather additional metadata, or less commonly, tags, which are user-generated keywords that can be added to any observation by the observer (see [Supplemental Text S3](#) for more details on these fields).

Workflows, and their tradeoffs, for accessing iNaturalist data

Researchers should select a data-access workflow based on the data required for their research question while maximizing reproducibility whenever possible ([figure 4](#)). We recommend using the simplest workflow that provides the necessary information, with GBIF serving as the preferred access point whenever it contains the required data. Detailed tutorials for each workflow, including example queries and code, are provided in [Supplemental Text S11](#).

For most biodiversity studies, we recommend downloading iNaturalist data through GBIF (<https://www.gbif.org/>), which aggregates all Research Grade observations released under CC0, CC BY, or CC BY-NC licenses. In addition to providing standardized Darwin Core occurrence records, GBIF assigns a DOI to every downloaded dataset, creating a permanent, citable snapshot of the exact data used in an analysis ([Supplemental Text S4](#)). This greatly simplifies reproducibility and provides a transparent chain of attribution back to iNaturalist contributors. Even when only a relatively small number of observa-

tions are required, we generally recommend downloading through GBIF whenever the necessary records and metadata are available. For studies requiring only one or a few individual observations, directly citing the corresponding iNaturalist observation URL or observation ID is generally sufficient.

When data unavailable through GBIF are required (e.g., Needs ID observations, observation fields, collection-project information, trusted coordinates, observations that are not licensed under CC0, CC BY, or CC BY-NC, or complete identification histories), we recommend using the iNaturalist Export Tool ([Supplemental Text S5–S6](#)). This web-based interface supports more complex filtering and provides access to additional metadata not included in GBIF. However, because Export Tool downloads do not generate versioned datasets or DOIs, researchers should carefully document download dates, search parameters, and archive the final dataset used for analysis.

The iNaturalist API should generally be reserved for situations where the required filters or returned data are unavailable through either GBIF or the Export Tool. Examples include specialized filters such as observations updated since a specified date, observations lacking taxonomic identifications, or excluding certain users from observation sets, as well as programmatic access to annotations and other nested metadata. This method provides the most granular control over both search filters and returned fields. For a full list of parameters, refer to the [iNaturalist API](#) documentation (iNaturalist 2026c; see [Supplemental Text S7](#) on how to modify request URLs for the API). Additionally, API calls by default return data in JSON format, which allows complex fields—such as media, annotations, and identifications—to be represented as nested data structures containing all relevant items and their associated metadata. Although the API offers the greatest flexibility, it also requires the greatest level of technical expertise and places the burden of reproducibility on the researcher. Accordingly, API users should archive derived datasets and report all endpoints, parameters, and access dates. Because the API is considerably more complex than the other access pathways, we provide detailed guidance, annotated example workflows in R and Python, and additional API resources in [Supplemental Texts S7–S10](#).

Regardless of the access route used, researchers should clearly document how data were obtained, report download dates and filters, and archive derived datasets where appropriate. These recommendations are expanded further in the Attribution and reproducibility section.

Best practices for analyzing iNaturalist data

iNaturalist is a large unstructured biodiversity dataset, created by observers who choose when, where, and what to record (Bowler et al. 2022). iNaturalist data are powerful for documenting spatiotemporal patterns of species occurrence, yet because the data are unstructured and heterogeneous across space, time, taxa, and observers, many of the underlying sources of heterogeneity are unreported (Geurts et al. 2023a). Scientific use requires understanding the data-generating process, matching analytical tools to the type of information the platform can provide, and applying careful filtering, modeling, and drawing appropriate inferences. In this section, we outline six overarching best practices, or topics of consideration, which range from technical decisions (e.g., filtering, accuracy thresholds) to conceptual decisions (e.g., scales of inference) to maximize the scientific use of iNaturalist data.

Table 1. Glossary of key terms used throughout this manuscript.

Term	Definition	Notes/implications for research use
Participatory Science	Engagement of the general public in the scientific endeavor. This approach is also referred to as <i>citizen science</i> or <i>community science</i> , although <i>participatory science</i> has become increasingly common because it is viewed as more inclusive. Additionally, <i>community science</i> may refer specifically to scientific efforts that are led by communities	Although we use the term participatory science throughout this paper, the scientific literature also uses the terms citizen science or community science
Observation	A record submitted to iNaturalist. Records that include media (photo/audio) and associated metadata (date, location) are deemed “verifiable”	Observations do not imply systematic sampling; absence of an observation of a species $\neq$ true absence of a species
Computer Vision	A machine-learning model that generates initial species identification suggestions from the user’s uploaded photos	Although initial identifications from this model can increase the likelihood that an observation reaches Research Grade, they should not be relied upon until they receive community support
Identification (ID)	A taxonomic label suggested by a user or computer vision for an observation. Multiple IDs can be attached to a single observation	Each ID is weighted equally, regardless of who suggested it
Community Taxon	The taxonomic consensus resulting from the community identification process based on a $>2/3$ majority agreement rule	IDs can change over time as more users review; the <i>community taxon</i> reflects consensus at a given point in time
Research Grade	A status assigned to observations that meet (1) verifiable metadata (date, location, media), (2) community taxon finer than family, and (3) $\geq 2/3$ ID agreement	Only Research Grade observations with appropriate licenses are exported to GBIF
Needs ID	Observations that are verifiable but have not yet reached community consensus at a fine taxonomic level	Often valuable in data-limited taxa; may include high-quality records missing expert review or only identified by a single expert
Casual	Observations lacking key metadata (e.g., no media or location), depicting captive/cultivated organisms, or having low-quality metadata (e.g., date is not accurate)	Usually excluded from biodiversity analyses unless the research question requires them
License	Usage rights chosen by observers for their observation data, images, and audio files (e.g., CC0, CC BY, CC BY-NC)	License type determines how data can be legally downloaded, shared, and exported to GBIF
Geoprivacy	Setting determining how precisely location data are displayed: <i>open</i> , <i>obscured</i> , or <i>private</i>	Obscured locations are randomized within a $0.2^\circ \times 0.2^\circ$ cell; private observations hide locations entirely and are not exported to GBIF
Obscured Observation	An observation whose coordinates are intentionally randomized to protect sensitive species or user privacy	Suitable for coarse-scale analyses; do not use for fine-scale habitat inference without buffers or without obtaining true coordinates
GBIF Export	The subset of iNaturalist observations and metadata shared automatically with GBIF under compatible licensing	Best for reproducibility because GBIF provides a DOI reference for citation
iNaturalist’s Export Tool	A web-based tool on iNaturalist that allows users to search for and download observation data	This method offers more filtering options and exports more data than GBIF, but queries are limited to 200,000 observations at a time, and DOI references are not provided
API Access	A programmatic method of requesting data directly from iNaturalist endpoints	Allows for full control and access to <i>all</i> metadata, but requires rate-limit awareness and documentation of query parameters
iNaturalist Open Data	Dataset of iNaturalist images, updated in real time, and their associated metadata, updated monthly. Hosted on Amazon Web Services	Photo URLs formatted as s3://inaturalist-open-data/photos or https://inaturalist-open-data.s3.amazonaws.com/photos are hosted on Amazon Web Services and can be downloaded without placing any load on the iNaturalist servers
Sampling Effort	The latent and uneven distribution of where, when, and what observers choose to record	Should be considered, and where relevant, modeled or otherwise accounted for to avoid misinterpreting sampling bias as ecological signal
Presence-only Data	Data describing where a species has been observed, without information on where it was searched for but <i>not</i> found	iNaturalist data are inherently presence-only; absences must be inferred statistically, never assumed
Reporting Rate	The fraction of all observations in a given unit that belong to a particular species	May approximate relative abundance if sampling effort is adequately modeled
Taxonomic Backbone	The taxonomic reference system used by iNaturalist to organize and reconcile species names	Changes over time; analysts should record the <i>date</i> when data were downloaded

We further refer the reader to this link with further terms relevant for iNaturalist: <https://www.inaturalist.org/terminology>.

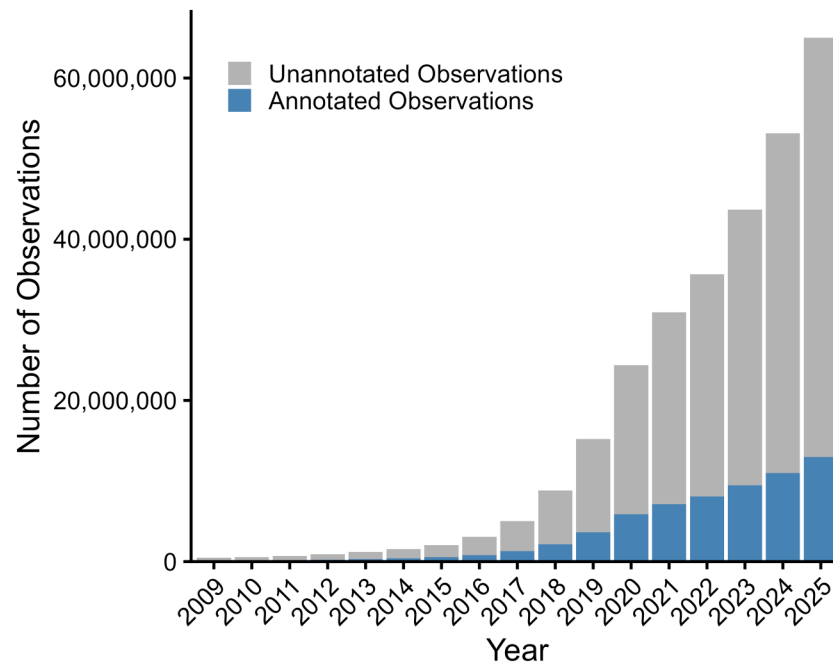


Figure 3. Number of observations with at least one annotation versus unannotated iNaturalist observations by year. Over 64 million observations made between 2009 and 2025 have at least one annotation.

Evaluate identification accuracy as necessary

iNaturalist species IDs are not perfectly accurate (e.g., McMullin and Allen 2022, Mesaglio et al. 2025). Even for easy-to-identify species, a small fraction of Research Grade observations may be misidentified, especially those based on indirect evidence (e.g., scat, footprints) rather than the organism itself. How much one must consider misidentifications should scale with the inferences being drawn. For example, if iNaturalist data are used for the detection of a threatened taxon or documenting novel range expansions, greater scrutiny should be placed on confirming the accuracy of the relevant observations. In contrast, a species distribution model fitted with tens of thousands of observations will not be sensitive to a handful of misidentified records (Graham et al. 2008), although sensitivity to false positives can vary widely depending on the statistical model (see, e.g., Clare et al. 2021, Van Eupen et al. 2021). In some cases, even using non-Research Grade data with some quality controls can provide useful information about biodiversity (Hochmair et al. 2020, Mesaglio et al. 2025). Because iNaturalist's identifications are dynamic and updated continuously, analysts should explicitly consider the potential for identification uncertainties and ensure that filtering choices (e.g., species-level versus genus-level identifications) align with the biological question.

Do not treat non-detections as absences

iNaturalist records are an extremely convenient source of information on species' presence. They are less suited, however, to establish the absence of a species. In areas with many iNaturalist records, a lack of observations of a common and readily identified species may strongly correspond to the absence of that species given that similar species are commonly reported in the area, but in areas of low overall iNaturalist effort or taxonomically specific iNaturalist effort, even common species may go unreported. Even the areas most saturated

with iNaturalist data do not document all species, and many that are low density, elusive, or especially diverse may simply not be detected. Conflating lack of iNaturalist data with absence is a dangerous but common mistake; in a review of iNaturalist data used in environmental impact reports, Callaghan and colleagues (2025) identified 28 cases where the lack of observations of a certain species was used to establish the absence of that species. We recommend against the use of iNaturalist data to provide evidence of species absence without the use of statistical analyses or supplemental professional sampling.

Apply question-specific filtering and quality control

Data filtering must match the research objective. When assessing or analyzing location data of iNaturalist observations, careful attention must be paid to whether coordinates have been obscured; otherwise, spurious conclusions may be drawn based on the randomized rather than the true coordinates of each record. For some questions, only Research Grade observations may be appropriate; for others, Needs ID records may contain valuable information, especially in data-limited taxa. In some cases, non-Research Grade observations do not have significantly worse taxonomic accuracy than Research Grade observations (Hochmair et al. 2020, Mesaglio et al. 2025). Similarly, geographic accuracy, date completeness, time stamps, annotations, and media requirements should be chosen intentionally. Filtering decisions should be explicitly justified in the context of the biological question, rather than applied as generic defaults. Choices about which fields to require, which taxonomic ranks to include, or how to treat incomplete meta-data directly shape the ecological inferences that can be drawn.

Match data treatment to the scale of inference

Across methodologies and taxa, analyses of iNaturalist data at very fine spatial scales can be especially challenging. Firstly, the geolocations associated with many iNaturalist observations have a level

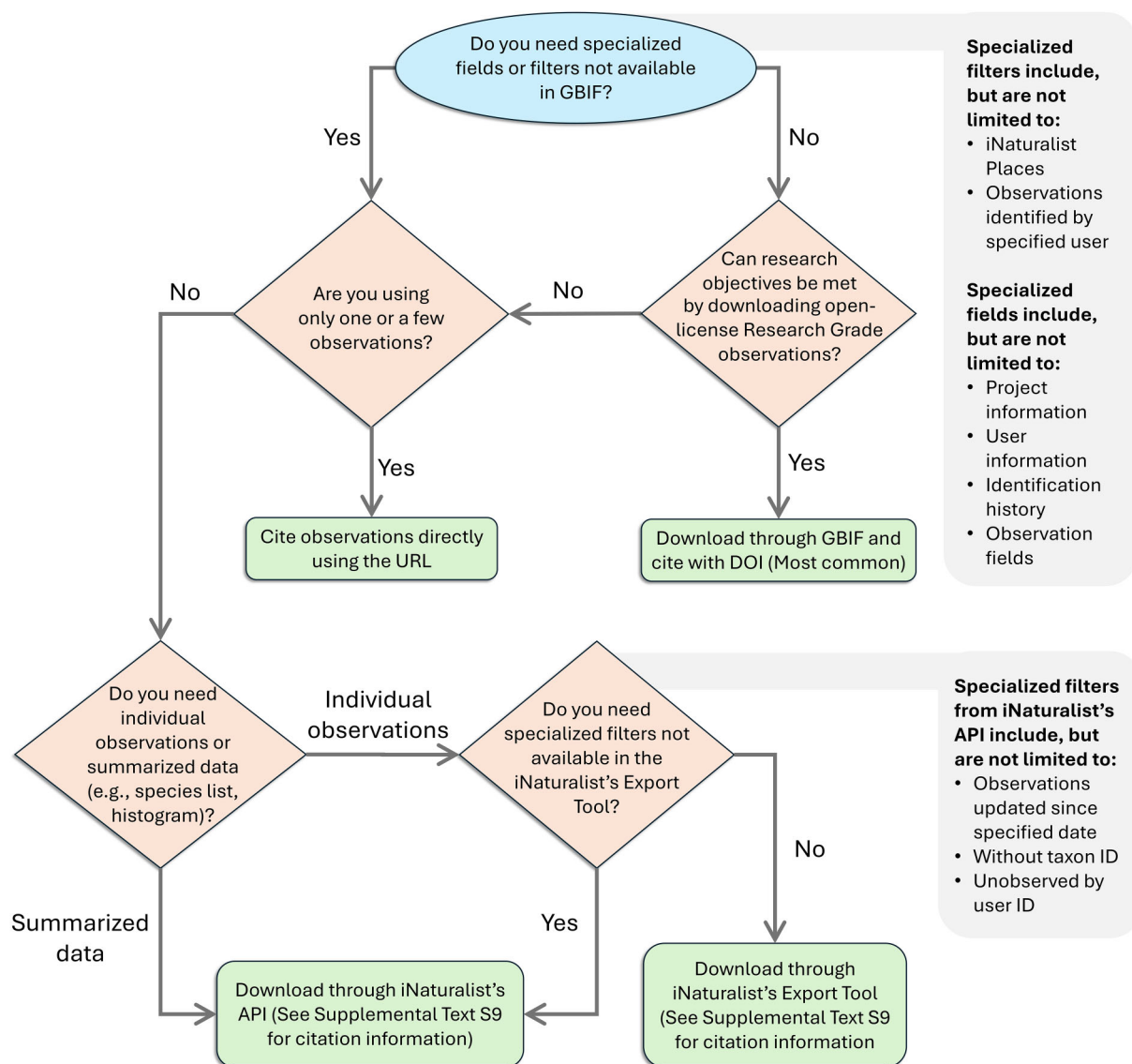


Figure 4. Flow chart to determine the best method of downloading iNaturalist observation data.

of uncertainty that is not conducive to very fine-scale applications (~10% of all Research Grade observations have a coordinate uncertainty of more than 500 m). Secondly, spatial variation in survey effort (where observers are not looking) becomes extreme at fine scales. Consequently, special care should be taken if modeling iNaturalist data at fine spatial scales (e.g., <1 km). The appropriate resolution will depend on the research question, taxon, sampling intensity, and treatment of positional uncertainty. Further, obscured records require special attention when considering scale of analyses, as obscuration replaces true coordinates with randomized coordinates within a $0.2^\circ \times 0.2^\circ$ grid cell. Researchers should determine whether these randomized coordinates are compatible with the resolution of their study, and when they are not, transparently report how these records were treated or excluded. Finally, if users filter observations by a specified threshold of positional uncertainty, they must also decide how to treat “NA” coordinate uncertainty values, which indicate missing information rather than necessarily high positional uncertainty.

The appropriate approach for dealing with record-level spatial uncertainty depends on the research objective. For analyses at coarse spatial scales, such as summaries of species occurrences across large regions (e.g., a state, province, county, etc.), coarse-grain data may still be useful at higher levels of uncertainty. However, for fine-scale analyses—for example, determining habitat associations or occurrences in small urban parks—obscured locations and locations with high uncertainty require careful treatment. In many cases, especially at fine spatial resolutions, records with obscured coordinates should not be used unless true coordinates can be assessed as they distort the basic information and analysis results (Koo et al. 2025). Typical approaches include filtering observations by a required level of location certainty, but in some cases, it may be reasonable to instead add buffers equal to the uncertainty radius around points and assess whether they overlap the area of interest. Analysts should be familiar with how positional uncertainty is determined in iNaturalist (see Supplemental Text S1).

Recognize and account for the heterogeneity of sampling

Users of iNaturalist data must contend with the fact that observation patterns arise from the behavior, preferences, and constraints of human observers (Grady et al. 2026). iNaturalist sampling is unstructured: observers post at their own discretion and typically do not report any metadata describing their search effort or motivations. Consequently, iNaturalist data arise from a sampling process that is uneven in space, time, and taxonomy. Thankfully, researchers have done a great deal of work characterizing these hidden processes. At the simplest level, iNaturalist sampling effort is clustered in times and places of high human activity, for example, near human population centers and along roads and trails (Geurts et al. 2023a, 2023b). Most observations are diurnal, and observer activity peaks in the spring and summer (though there is substantial geographic variation in seasonal patterns; Di Cecco et al. 2021, Forti and Szabo 2025). Such heterogeneity in sampling effort is compounded by mass-scale organized participation events (e.g., City Nature Challenge, Moth Week, Great Southern Bioblitz), which can create pronounced spatial and temporal pulses in observation effort that may influence analyses if left unaccounted for (Di Cecco et al. 2021). Observation effort also varies across taxa. Observers tend to preferentially report larger, rarer, and more colorful species (Callaghan et al. 2021, Stoudt et al. 2022). Additionally, because verifiable iNaturalist observations require evidence of the organism through photo or sound recording, there is a bias toward species that are easier to document based on size, behavior, activity patterns, and/or volume of their call (Arazy and Milkson 2021, Di Cecco et al. 2021, Forti and Szabo 2025). Individual observers also specialize in particular taxa and repeatedly observe them, meaning that search effort may be spatially clustered (Di Cecco et al. 2021, Goldstein and Stoudt 2025). This structured heterogeneity matters because it shapes detection probabilities, can bias richness estimates, and complicates comparisons across sites or taxa. Any use of iNaturalist data should consider this uneven sampling effort and ensure that the scientific question is robust to it or that these patterns are accounted for statistically.

A variety of specific modeling approaches to account for heterogeneity in iNaturalist data have been demonstrated in the literature. Many researchers retrieve and visualize iNaturalist data simply to establish places where the species occurs with certainty, then use critical thinking and interpretation of auxiliary data to weigh the roles of sampling effort and species absence in causing spatial gaps in species detections (e.g., Hobson et al. 2017, Jachowski et al. 2021). This approach is reasonable, especially for hard-to-detect species where iNaturalist data are often the best available information on a species' range and for applications where statistical inference of spatial effects is not of interest. Researchers may also vary their spatial scale of inferences so that the influence of heterogeneity is minimized. For instance, researchers found that the occurrence of common species was better documented at the county level as opposed to the town level (Bertin and Grima 2026). Another common approach is to thin iNaturalist data to standardize effort before modeling occurrences as presence-only data under the assumption that effort is constant (e.g., Dimson et al. 2023, Hill et al. 2024). For example, an analyst may retain only one observation of the focal taxon (or focal taxonomic group) per 10-km spatial grid cell. This approach may be appropriate for some cases, but it breaks down when sampling effort varies dramatically at the scale of spatial control. Furthermore, emerging analytical approaches

are improving the ability to use presence-only data to more accurately estimate species distribution models (Valavi et al. 2022). Researchers may also account for missing data by utilizing covariates and knowledge of the system in statistical workflows (Bowler et al. 2025). Finally, some approaches generate non-detection data to represent heterogeneous sampling (see the "Estimating non-detections" section).

Make clear inferences in the context of how iNaturalist operates

Many observer tendencies are consistent *across* broad gradients—people often observe similar kinds of organisms in similar ways in urban, suburban, and rural settings—but the *intensity* of observation effort can vary dramatically along these same gradients (Di Cecco et al. 2021). This means that some questions (e.g., coarse-scale range limits or broad biogeographic patterns) are robust to uneven effort, whereas others (e.g., local abundance or fine-scale habitat preferences) are far more sensitive to it. For instance, research comparing reptile and amphibian occurrence data from iNaturalist with data from traditional surveys found that iNaturalist observations were useful for estimating species richness but were not sufficiently robust for analyzing community structure or diversity without statistical techniques to correct for sampling bias (Marsh et al. 2026). Similarly, researchers evaluating the use of iNaturalist to characterize a mass mortality event caused by avian influenza found that the platform could identify the species affected, as well as the timing and location of mortality events, but could not reliably quantify the magnitude or underlying mechanisms of mortality (Taylor et al. 2024). Researchers should therefore interpret results conservatively when appropriate, articulate how sampling processes may interact with their hypotheses, and transparently report the assumptions and uncertainties underlying their inferences.

Attribution and reproducibility in published research using iNaturalist data

One of the many strengths of iNaturalist data is that they are "living," meaning that almost any aspect, including their existence, could change between iterative downloads. This dynamism allows for the improvement of taxonomic identifications, addition of data quality annotations, and correction of errors, but it can also result in meaningful changes to existing iNaturalist data at any time. This might mean that observations are deleted; or their licensing, taxonomic identification, geographic associations, or other attributes changed. It also means that analyses are not always exactly replicable unless the originally used data are available. At present, articles that do not provide the versioned dataset used for analysis often lack sufficient information for the dataset to be reconstructed (Supplemental Text S13). To account for this, we suggest that at a minimum, researchers explicitly document how and when the data were accessed.

As discussed above in detail, GBIF-mediated downloads generate versioned, citable datasets that preserve the state of iNaturalist records at the time of download for at least 6 months and often longer (see Supplemental Text S8 for further details). In contrast, data obtained through the iNaturalist API or export tool require users to manage their own versioning, including retaining downloaded data locally and documenting all filters, endpoints, and access dates, which should be reported and end products archived. These key data man-

agement aspects are essential for scientific transparency and compliance with FAIR (Findable, Accessible, Interoperable, and Reusable) principles, which are increasingly recognized for their importance by journals, funders, and researchers in ecology and evolution research (Manzano and Julier 2021, Koivisto and Mäntylä 2024). Although FAIR principles should guide data, metadata, and other digital objects (e.g., code) created by researchers, community science datasets also require respect for contributor decisions about licensing and CARE principles for contributor-generated data (Jennings et al. 2023).

Below, we provide a concise checklist of information that should be included when publishing analyses using iNaturalist data. These recommendations are intended to improve reproducibility while recognizing that iNaturalist datasets are generated through the collective efforts of a global community of observers and identifiers whose contributions underpin biodiversity research. Following these steps will maximize reproducibility, ensure proper attribution to iNaturalist contributors, that is, observers and identifiers, and support the long-term integrity of biodiversity data workflows (see [Supplemental Text S14](#) for reporting templates).

1. Include the pathway to data access.

- (a) Report GBIF download DOI(s), including the exact query used.
- (b) Report iNaturalist API endpoints or Export Tool URLs used, including all parameters.
- (c) State whether multiple data sources were combined (e.g., GBIF + API).

2. Specify the exact filters applied.

- (a) Specify quality grades included (Research Grade, Needs ID, Casual).
- (b) Specify taxonomic, geographic, temporal, annotation, and observation field filters.
- (c) Report any user-level filters applied (e.g., exclusion lists, project-based queries).

3. Document download and access dates.

- (a) Report the date(s) of data retrieval.
- (b) Specify any “created on” query used to define a terminal inclusion date prior to download.

4. Report licensing and usage conditions.

- (a) Summarize the licenses attributed to observations and associated media.
- (b) Describe how licensing restrictions were enforced (e.g., exclusion of “all rights reserved” content).
- (c) Confirm compliance with all applicable license terms.

5. State taxonomic versioning details.

- (a) Report the date of taxonomic tree or backbone retrieval, if used.
- (b) Describe how taxonomic changes (splits, lumps, renames) were handled.
- (c) Specify whether iNaturalist or GBIF taxonomic backbones were used.

6. Include geoprivacy and coordinate resolution handling.

- (a) Describe how obscured and/or private observations were treated.

- (b) Specify thresholds used for coordinate uncertainty filtering.
- (c) Report any buffers or uncertainty-based exclusion rules applied.

7. Archive data in a permanent repository.

- (a) Specify where processed datasets are archived (e.g., Zenodo, GitHub, Dryad).
- (b) Archive all observation IDs used in analyses where permissible, noting that trusted coordinates cannot be re-shared and that archived datasets may therefore be incomplete.

Emerging research frontiers for iNaturalist data

iNaturalist data have seen a tenfold increase in use in scientific literature between 2017 and 2022 (Mason et al. 2025), illustrating the vast potential of these data for use in biodiversity research and conservation. Nevertheless, as described above, these data are complex, heterogeneous along multiple dimensions, and therefore require careful treatment in analyses. The challenge moving forward is not whether iNaturalist data *can* be used, but *how best to use them* and maximize the value of these data for biodiversity and conservation research while avoiding erroneous interpretations. Here, we highlight four different forward-looking uses of iNaturalist data, which we feel represent the next wave of potential research questions.

Secondary data

Although iNaturalist data usage has primarily been focused on the spatiotemporal coordinates of observations, an increasing number of papers extract ecological information from images (Mason et al. 2025). Such “secondary data” (Pernat et al. 2024a), defined as additional information unintentionally captured in species records, are becoming an important and expanding research frontier. Photos from iNaturalist have been used to investigate plant-pollinator interactions (e.g., Pernat et al. 2024b), obtain color data of flowers (Luong et al. 2023), quantify morphological traits such as plumage patterns in birds (Vretos et al. 2021) or color polymorphisms in salamanders (Hantak et al. 2022), study dietary patterns in birds (Panter et al. 2024), identify mating behavior (Ament and Brown 2025), and assess habitat composition (Bolt et al. 2022), as just a few examples. Another example within the frontier of secondary data is the field of phenology, where image timestamps allow flowering times and leaf-outs (Dinnage et al. 2025) as well as animal diel activity patterns (Idec et al. 2025) to be quantified at large geographic and taxonomic scales. This growing field of secondary data use will continue to shape how iNaturalist data are used in the future. As large-scale text-to-image retrieval and multimodal machine learning systems continue to advance (Vendrow et al. 2024), we expect secondary-data applications to play a central role in the next wave of biodiversity research that uses iNaturalist.

Data integration

As researchers extract increasingly diverse ecological information from iNaturalist records, a natural next step is to combine these rich, heterogeneous data with other biodiversity datasets. An increasingly important research frontier is incorporating iNaturalist data with standardized data (e.g., structured or semi-structured survey data or specimen data from natural history collections) to better inform ecological and biogeographic studies (e.g., Yoder et al. 2024), taxonomic

works (e.g., Smith et al. 2022), or conservation and management initiatives (e.g., Ackerfield et al. 2024). In many cases, data integration maximizes the temporal and spatial breadth of the data and improves our understanding of biodiversity from local to global scales (Mesaglio 2025). Although the number and diversity of research papers using integrated datasets have rapidly increased over the last decade (Mason et al. 2025), there is still broad scope and exciting opportunities to further integrate datasets.

Understanding the best approaches and methods for integrating iNaturalist data and traditional data sources, and when or where these approaches are applicable, is critically important. Here, data integration refers not simply to combining datasets, but to analytical frameworks that explicitly account for the differing sampling designs, biases, and uncertainties associated with multiple data sources (Isaac et al. 2020, Mostert and O'Hara 2023). This can be done at both the scale of entire datasets and through more tailored approaches involving individual records or curated partial datasets. These principles may also apply within iNaturalist itself. For example, observations originating from structured projects, coordinated participation events, or different contributor groups may arise from distinct sampling processes that could warrant separate consideration depending on the research question. A key frontier is understanding how integrative approaches can mitigate the idiosyncratic biases present in each data stream. By combining datasets with different strengths and weaknesses, researchers may be able to reduce bias, increase completeness, and better capture true ecological and biogeographic patterns (see Calvente et al. 2023 for a compelling recent example). Developing curated frameworks for combining iNaturalist data with structured surveys, museum collections, and emerging sensor-based datasets will be essential for fully realizing the scientific potential of biodiversity data integration and unlocking the full potential of iNaturalist data.

Estimating non-detections

Even with integrated datasets, many analyses require non-detection information, which has fueled a growing body of research focused on estimating pseudo-absences or non-detections from unstructured biodiversity data (Rapacciuolo et al. 2021, Boyd et al. 2023, Davis et al. 2023, Morera-Pujol et al. 2023, Bowler et al. 2025, van Strien et al. 2026). Because iNaturalist is inherently a presence-only dataset, some analyses and inferences are limited (e.g., Guillera-Arroita et al. 2015), and it means estimating detectability is more difficult than with presence-absence datasets. A promising frontier of current and future research is the development of approaches that derive and/or estimate non-detection information from unstructured iNaturalist data. One direction is the construction of participant-level sampling events, where observations within a place-time window are grouped into a "visit" or "checklist." Rapacciuolo and colleagues (2021) proposed a framework for deriving participant events (akin to checklists in eBird), which could be used to demarcate everything a participant could have documented in a sampling event. For example, when participants blacklight to attract moths, inferences could be made about the species that were not observed during that blacklighting event. Another strategy leverages trait-based or ecological similarity to infer plausible pseudo-absences (e.g., Koen et al. 2025, Sibiya et al. 2025). Species with comparable detectability, body size, microhabitat use, or phenology can be used to estimate when a focal species might rea-

sonably have been observed but was not, thereby providing estimates of non-detections. Best practices for deriving non-detections from unstructured data remain an open challenge (Davis et al. 2023), and future research should focus on clarifying when different strategies are appropriate and how they influence downstream inference.

Trend detection

Estimating biodiversity trends, either at a community level, or a species-specific level, remains an important goal for biodiversity monitoring and research. iNaturalist was launched in 2008, and with increasing time, there is an increasing amount of data to potentially estimate trends. A key frontier for iNaturalist research, therefore, is determining if, when, and how these data can be used to infer occurrence or abundance trends. A fundamental research question is whether the frequency of iNaturalist records in a region approximates the relative abundance of the species in that region. If a stable, interpretable relationship exists, then trend detection using iNaturalist data may become statistically feasible. For instance, if a species is consistently observed hundreds of times per year and then suddenly disappears from new contributions to the platform, would this be driven by a true biological signal, a change in observer behavior, or a consequence of sampling heterogeneity? Disentangling these potential relationships and determining how reliably trends can be identified across space and time are critical open questions for the use of iNaturalist data in biodiversity monitoring.

Emerging research is starting to address these questions. For example, Kaiser and colleagues (2026) used a synthetic-control framework to tease apart causal patterns in iNaturalist reporting, showing how a disturbance event influenced bee detections while appropriately accounting for uneven spatial sampling across regions. Additionally, researchers aiming to make use of iNaturalist data to characterize variation in species' relative abundance, and conduct inference on the drivers of said variation have introduced approaches that attempt to address the problem of spatial variation in effort through model-based inference (Soberón and Christén 2026). Some approaches assume that the reporting rate of a species on iNaturalist is a function of its relative abundance (Ringim et al. 2026), which is reasonable in many cases but breaks down if the abundance of all target species strongly covaries or if the target species makes up a large percentage of all reports in an area of interest.

To fully disentangle biological trends from sampling, research needs to continue to (1) quantify and understand observer effort and reporting behavior, (2) integrate iNaturalist data with structured survey data, and (3) develop and refine statistical approaches suitable for unstructured presence-only time series data for both occupancy and especially relative abundance (Rapacciuolo et al. 2021). Importantly, progress in trend detection will likely depend on developments across several of the other frontiers outlined above.

Conclusions

iNaturalist has become one of the world's most important sources of biodiversity information, and its data are sometimes held to a higher standard than professionally collected datasets despite both arising from imperfect sampling processes (Binley and Bennett 2023). Their scientific value depends on researchers understanding how the data are generated, accessed, filtered, analyzed, and attributed. Through-

out this guide, we highlight best practices that maximize reproducibility, ensure appropriate interpretation of observations, and match analytical approaches to the strengths and limitations of this dynamic dataset. As with any observational dataset, the responsibility for appropriate analysis lies with the researcher. Researchers should explicitly consider the assumptions, biases, and limitations inherent in iNaturalist and ensure that their scientific inferences are appropriate to those characteristics.

Reciprocity and community stewardship

iNaturalist should not be viewed only as a data source: this best-practices-oriented review also focuses on its roots as an evidence-oriented community participatory platform. The scientific value of iNaturalist is built on the enthusiasm of millions of people who choose to share their time, curiosity, and observations. It is essential that researchers approach iNaturalist as a collaborative knowledge-making and sharing system that also supports the betterment of biodiversity research. Respecting licenses and contributor intent is the minimum standard for any use. Wherever possible, researchers should also work to give back to the community. Researchers can credit observers and identifiers in publications, offer co-authorship to significant contributors, share project results in accessible forms, acknowledge taxonomic expertise, and participate in the platform by identifying, annotating, or engaging with users (Callaghan et al. 2022). Such acts of reciprocal contribution strengthen trust, improve data quality, sustain long-term participation, and grow the community. As biodiversity research continues to leverage participatory science data, such principles will help ensure the community builds the richest possible network of data, people, and evidence to support biodiversity research.

Example code and workflows

To help facilitate broad use of iNaturalist data, we have made example code available at <https://github.com/Global-Ecology-Research-Group/Guidelines-and-Best-Practices-for-iNaturalist-Data>. This repository currently contains example R and Python scripts for common ways to access and work with iNaturalist data:

1. **Get Observations:** Code to get observations from a URL request. For example, all butterfly observations in Texas.
2. **Species Counts:** Code to get a table of species counts from one URL request. For example, get a count of observations of all angiosperm species observed in Florida.
3. **Download Images from URL:** Code to download iNaturalist images from a URL path. In the example, observation data are gathered from the iNaturalist API. Then parallel processing is used to download images hosted on Amazon Web Services. Images are named according to observation ID and photo number, so they can be linked to the full observation details later.
4. **Annotation Data:** Code to get annotation data from an API request.
5. **Identifier Data:** Code to get information on identifiers from an API request.
6. **Observer Data:** Code to get information on observers from an API request.
7. **Popular Field Values:** Code to get relevant controlled term values and monthly histogram of observations.

Author contributions

Corey T. Callaghan (Conceptualization, Investigation, Methodology, Project administration, Resources, Software, Supervision, Visualization, Writing – original draft, Writing – review & editing), C.J. Campbell (Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing), Jane Widness (Conceptualization, Data curation, Formal Analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing), Benjamin R. Goldstein (Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing), Tony Iwane (Investigation, Resources, Supervision, Validation, Visualization), Thomas Mesaglio (Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing), Carrie Seltzer (Conceptualization, Resources, Software, Supervision, Writing – original draft, Writing – review & editing), Robert Guralnick (Investigation, Methodology, Resources, Validation, Writing – original draft, Writing – review & editing), Brittany M. Mason (Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Visualization, Writing – original draft, Writing – review & editing)

Supplementary data

Supplementary data are available at *BIOSCI* online.

References cited

- Ackerfield J, Boom BM, Gandy E, Paradiso L. EcoFloras elucidate insights from biodiversity data: Evaluating the strengths and limitations of iNaturalist observations and herbarium specimens. 2024. *Preprints.org*. <https://doi.org/10.20944/preprints202410.0017.v1>
- Ament DC, Brown BV. 2025. Phoridae (Diptera) mating behaviour: Identification of general patterns, exceptions and the use of sexually dimorphic structures based on vast citizen science data. *Journal of Natural History* 59: 739–761.
- Arazy O, Malkinson D. (2021). A framework of observer-based biases in citizen science biodiversity monitoring: Semi-structuring unstructured biodiversity monitoring protocols. *Frontiers in Ecology and Evolution* 9: 693602. <https://doi.org/10.3389/fevo.2021.693602>
- Bánki O et al. 2026. *Catalogue of Life (2026-01-16 XR)*. Catalogue of Life Foundation. <https://doi.org/10.48580/dgw64>
- Beninde J, Delaney TW, Gonzalez G, Shaffer HB. 2023. Harnessing iNaturalist to quantify hotspots of urban biodiversity: The Los Angeles case study. *Frontiers in Ecology and Evolution* 11: 983371.
- Bertin RI, Grima PP. 2026. Coverage and biases in iNaturalist records of vascular plants in Massachusetts. *Rhodora* 126: 351–379.
- Binley AD, Bennett JR. (2023). The data double standard. *Methods in Ecology and Evolution* 14: 1389–1397.
- Bolt MH, Callaghan CT, Poore AG, Vergés A, Roberts CJ. 2022. Using the background of fish photographs to quantify habitat composition in marine ecosystems. *Marine Ecology Progress Series* 688: 167–172.
- Bowler DE et al. 2022. Decision-making of citizen scientists when recording species observations. *Scientific Reports* 12: 11069.
- Bowler DE, Boyd RJ, Callaghan CT, Robinson RA, Isaac NJ, Pocock MJ. (2025). Treating gaps and biases in biodiversity data as a missing data problem. *Biological Reviews* 100: 50–67.
- Boyd RJ, August TA, Cooke R, Logie M, Mancini F, Powney GD, Isaac NJ. (2023). An operational workflow for producing periodic estimates of species occupancy at national scales. *Biological Reviews* 98: 1492–1508.

- Callaghan CT et al. 2022. The benefits of contributing to the citizen science platform iNaturalist as an identifier. *PLoS Biology* 20: e3001843.
- Callaghan CT, Poore AG, Mesaglio T, Moles AT, Nakagawa S, Roberts C, Rowley JLL, Vergés A, Wilshire JH, Cornwell WK. 2021. Three frontiers for the future of biodiversity research using citizen science data. *BioScience* 71: 55–63.
- Callaghan CT, Winnebald C, Smith B, Mason BM, López-Hoffman L. 2025. Citizen science as a valuable tool for environmental review. *Frontiers in Ecology and the Environment* 23: e2808.
- Calvente A, da Silva APA, Edler D, Carvalho FA, Fantinati MR, Zizka A, Antonelli A. 2023. Spiny but photogenic: Amateur sightings complement herbarium specimens to reveal the bioregions of cacti. *American Journal of Botany* 110: e16235.
- Campbell CJ, Barve V, Belitz MW, Doby JR, White E, Seltzer C, Di Cecco G, Hurlbert AH, Guralnick R. 2023. Identifying the identifiers: How iNaturalist facilitates collaborative, research-relevant data generation and why it matters for biodiversity science. *BioScience* 73: 533–541.
- Clare JD, Townsend PA, Zuckerberg B. 2021. Generalized model-based solutions to false-positive error in species detection/nondetection data. *Ecology* 102: e03241.
- Cull B. 2022. Monitoring trends in distribution and seasonality of medically important ticks in North America using online crowdsourced records from iNaturalist. *Insects* 13: 404.
- Davis CL, Guralnick RP, Zipkin EF. 2023. Challenges and opportunities for using natural history collections to estimate insect population trends. *Journal of Animal Ecology* 92: 237–249.
- Di Cecco GJ, Barve V, Belitz MW, Stucky BJ, Guralnick RP, Hurlbert AH. 2021. Observing the observers: How participants contribute data to iNaturalist and implications for biodiversity science. *BioScience* 71: 1179–1188.
- Dimson M, Berio Fortini L, Tingley MW, Gillespie TW. 2023. Citizen science can complement professional invasive plant surveys and improve estimates of suitable habitat. *Diversity and Distributions* 29: 1141–1156.
- Dinnage R, Grady E, Neal N, Deck J, Denny E, Walls R Seltzer C, Guralnick R, Li D. 2025. PhenoVision: A framework for automating and delivering research-ready plant phenology data from field images. *Methods in Ecology and Evolution* 16: 1763–1780.
- Feldman MJ, Imbeau L, Marchand P, Mazerolle MJ, Darveau M, Fenton NJ. 2021. Trends and gaps in the use of citizen science derived data as input for species distribution models: A quantitative review. *PLoS ONE* 16: e0234587.
- Forti LR, Hepp F, de Souza JM, Protazio A, Szabo JK. 2022. Climate drives anuran breeding phenology in a continental perspective as revealed by citizen-collected data. *Diversity and Distributions* 28: 2094–2109.
- Forti LR, Szabo JK. 2025. Global community science data on mammals underreport small and diurnal species. *Environmental Monitoring and Assessment* 197: 1251.
- Geurts EM, Reynolds JD, Starzomski BM. 2023a. Turning observations into biodiversity data: Broad-scale spatial biases in community science. *Ecosphere* 14: e4582.
- Geurts EM, Reynolds JD, Starzomski BM. 2023b. Not all who wander are lost: Trail bias in community science. *PLoS ONE* 18: e0287150.
- Goldstein BR, Stoudt S. 2025. Evidence of novelty and specialization behavior in participatory science reporting. *Oikos* 2025: e10938.
- Grady EL, Campbell CJ, Callaghan CT, Guralnick RP. 2026. iNaturalist users exhibit distinct spatiotemporal sampling preferences, with implications for biodiversity science and project planning. *Citizen Science: Theory and Practice* 11: 3.
- Graham CH, Elith J, Hijmans RJ, Guisan A, Townsend Peterson A, Loiselle BA, NCEAS Predicting Species Distributions Working Group. 2008. The influence of spatial errors in species occurrence data used in distribution models. *Journal of Applied Ecology* 45: 239–247.
- Guillera-Arroita G, Lahoz-Monfort JJ, Elith J, Gordon A, Kujala H, Lentini PE, McCarthy MA, Tingley R, Wintle BA. (2015). Is my species distribution model fit for purpose? Matching data and models to applications. *Global Ecology and Biogeography* 24: 276–292. <https://doi.org/10.1111/geb.12268>
- Hantak MM, Guralnick RP, Zare A, Stucky BJ. 2022. Computer vision for assessing species color pattern variation from web-based community science images. *Isis* 25: 104784.
- Heberling JM, Miller JT, Noesgaard D, Weingart SB, Schigel D. 2021. Data integration enables global biodiversity synthesis. *Proceedings of the National Academy of Sciences* 118: e2018093118.
- Hill J, Teal E, Cross CL, Sanchez Z, Webber MM, Oxborough RM, Messenger LA. 2024. Using iNaturalist presence data to produce suitability maps for *Triatoma protracta*, *T. rubida* and *T. recurva* in the American Southwest, Texas and northern Mexico, to identify potential transmission zones of Chagas disease. *Scientific Reports* 14: 26879.
- Hobson EA, Smith-Vidaurre G, Salinas-Melgoza A. 2017. History of non-native monk parakeets in Mexico. *PLoS ONE* 12: e0184771.
- Hochmair HH, Scheffrahn RH, Basille M, Boone M. 2020. Evaluating the data quality of iNaturalist termite records. *PLoS ONE* 15: e0226534.
- Idec J, Campbell CJ, Belitz M, Akshay VA, Guralnick R. 2025. Using citizen science data to estimate trait and climate drivers of daily activity patterns in temperate butterflies. *PLoS ONE* 20: e0335856.
- iNatHelp. 2023. How can I access the hidden coordinates of sensitive species?. (11 March 2026; <https://help.inaturalist.org/en/support/solutions/articles/151000170343-how-can-i-access-the-hidden-coordinates-of-sensitive-species->)
- iNatHelp. 2024. What are licenses? How can I update the licenses on my content?. (11 March 2026; <https://help.inaturalist.org/en/support/solutions/articles/151000175695-what-are-licenses-how-can-i-update-the-licenses-on-my-content->)
- iNaturalist. 2026a. iNaturalist Network. (11 March 2026; www.inaturalist.org/sites/network)
- iNaturalist. 2026b. iNaturalist Open Range Map Dataset 2.28. (11 March 2026; www.inaturalist.org)
- iNaturalist. 2026c. iNaturalist API v2 documentation. (11 March 2026; <https://api.inaturalist.org/v2/docs/>)
- Isaac NJ et al. 2020. Data integration for large-scale models of species distributions. *Trends in Ecology & Evolution* 35: 56–67.
- Jachowski D, Kays R, Butler A, Hoylman AM, Gompfer ME. 2021. Tracking the decline of weasels in North America. *PLoS ONE* 16: e0254387.
- Jennings L, Anderson T, Martinez A, Sterling R, Chavez DD, Garba I, Hudson M, Garrison NA, Carroll SR. 2023. Applying the CARE Principles for Indigenous Data Governance to ecology and biodiversity research. *Nature Ecology & Evolution* 7: 1547–1551.
- Johnston A, Hochachka WM, Strimas-Mackey ME, Ruiz Gutierrez V, Robinson OJ, Miller ET, Auer T, Kelling ST, Fink D. 2021. Analytical guidelines to increase the value of community science data: An example using eBird data to estimate species distributions. *Diversity and Distributions* 27: 1265–1277.
- Kaiser A, Dee L, Resasco J. 2026. Synthetic control methods enable stronger causal inference using participatory science data in cities. *Nature Ecology and Evolution* 10: 1204–1215. <https://doi.org/10.1038/s41559-026-03084-4>

- Koen EL, Ellington EH, Barichivich WJ, Kochman H, Enge KM, Walls SC. 2025. Persistence of a declining anuran species across its distribution. *PLoS ONE* 20: e0332991.
- Koivisto E, Mäntylä E. 2024. Are open science instructions targeted to ecologists and evolutionary biologists sufficient? A literature review of guidelines and journal data policies. *Ecology and Evolution* 14: e11698.
- Koo KS, Lee KH, Lee D, Jang Y. 2025. Impact of obscured data on species distribution models. *Conservation Biology* 39: e70050.
- Luong Y, Gasca-Herrera A, Misiewicz TM, Carter BE. 2023. A pipeline for the rapid collection of color data from photographs. *Applications in Plant Sciences* 11: e11546.
- McMullin RT, Allen JL. 2022. An assessment of data accuracy and best practice recommendations for observations of lichens and other taxonomically difficult taxa on iNaturalist. *Botany* 100: 491–497.
- Manzano S, Julier A. 2021. How FAIR are plant sciences in the twenty-first century? The pressing need for reproducibility in plant ecology and evolution. *Proceedings of the Royal Society B: Biological Sciences* 288.
- Marsh DM, Finnegan T, Holmes C, Li J. 2026. Comparing iNaturalist and traditional surveys for estimating richness, diversity, and abundance of reptiles and amphibians. *Herpetologica* 82: 1–11.
- Mason BM et al. 2025. iNaturalist accelerates biodiversity research. *BioScience* 75: 953–965.
- Mesaglio T. 2025. *Reimagining the Botanical Biodiversity Informatics Landscape: Integrating Physical Vouchers and Photographic Vouchers Maximises Our Knowledge of the World's Flora*. Doctoral dissertation, University of New South Wales, Australia.
- Mesaglio T, Shepherd KA, Wege JA, Barrett RL, Sauquet H, Cornwell WK. 2025. Expert identification blitz: A rapid high value approach for assessing and improving iNaturalist identification accuracy and data precision and confidence. *Plants, People, Planet* 7: 1469–1484.
- Morera-Pujol V, Mostert PS, Murphy KJ, Burkitt T, Coad B, McMahon BJ, Ciuti S. 2023. Bayesian species distribution models integrate presence-only and presence-absence data to predict deer distribution and relative abundance. *Ecography* 2023: e06451.
- Mostert PS, O'Hara RB. 2023. PointedSDMs: An R package to help facilitate the construction of integrated species distribution models. *Methods in Ecology and Evolution* 14: 1200–1207.
- Natural History Museum Bern. 2026. World Spider Catalog (version 27). (11 March 2026; <https://wsc.nmbe.ch/>)
- Panter CT, Naude VN, Barbar F, Amar A. 2024. Continental scale dietary patterns in a New World raptor using web-sourced photographs. *PLoS ONE* 19: e0304740.
- Perez-Udell RA, Udell AT, Chang SM. 2023. An automated pipeline for supervised classification of petal color from citizen science photographs. *Applications in Plant Sciences* 11: e11505.
- Pernat N et al. 2024a. Overcoming biodiversity blindness: Secondary data in primary citizen science observations. *Ecological Solutions and Evidence* 5: e12295.
- Pernat N, Memedemin D, August T, Preda C, Reyserhove L, Schirmel J, Groom Q. 2024b. Extracting secondary data from citizen science images reveals host flower preferences of the Mexican grass-carrying wasp *Isodontia mexicana* in its native and introduced ranges. *Ecology and Evolution* 14: e11537.
- Rapacciuolo G, Young A, Johnson R. 2021. Deriving indicators of biodiversity change from unstructured community-contributed data. *Oikos* 130: 1225–1239.
- Ringim AS, Muhammad SI, Abubakar HM. (2026). Ecological traits as predictors of avian reporting rates and monitoring gaps: A 12-year analysis of iNaturalist in Nigeria. *Frontiers in Bird Science* 5: 1757541.
- Royal Botanic Gardens Kew. 2026. Plants of the World Online. (11 March 2026; <https://powo.science.kew.org/>)
- Sibiya MD et al. 2025. Assessing spatial ecological patterns of wild turkey relative abundance across North America. *Wildlife Society Bulletin* 49: e1633.
- Smith LT et al. 2022. Revised species delimitation in the giant water lily genus *Victoria* (Nymphaeaceae) confirms a new species and has implications for its conservation. *Frontiers in Plant Science* 13: 883151.
- Soberón J, Christén A. 2026. On using iNaturalist data to estimate trends. *Biodiversity Informatics* 20: 5–15.
- Stoudt S, Goldstein BR, de Valpine P. 2022. Identifying engaging bird species and traits with community science observations. *Proceedings of the National Academy of Sciences* 119: e2110156119.
- Strimas-Mackey M, Hochachka WM, Ruiz-Gutierrez V, Robinson OJ, Miller ET, Auer T, Kelling S, Fink D, Johnston A. 2023. *Best Practices for Using eBird data (version 2.0)*. Cornell Lab of Ornithology. (26 August 2026; <https://ebird.github.io/ebird-best-practices/>)
- Taylor LU, Barychka T, McKeon S, Bartolotta N, Avery-Gomm S. 2024. Strengths and limitations of using participatory science data to characterize a wildlife mass mortality event. *Ecosphere* 15: e70051.
- Valavi R, Guillera-Aroita G, Lahoz-Monfort JJ, Elith J. 2022. Predictive performance of presence-only species distribution models: A benchmark study with reproducible code. *Ecological Monographs* 92: e01486.
- Van Eupen C, Maes D, Herremans M, Swinnen KR, Somers B, Luca S. 2021. The impact of data quality filtering of opportunistic citizen science data on species distribution model performance. *Ecological Modelling* 444: 109453.
- van Strien AJ, Meijaard E, Nardiyono, Suief S, Zaini S. 2026. Occupancy estimation of wild species in a palm oil plantation using unstructured data. *PLoS ONE* 21: e0328960.
- Vendrow E, Pantazis O, Shepard A, Brostow G, Jones KE, Mac Aodha O, Beery S, Van Horn G. 2024. INQUIRE: A natural world text-to-image retrieval benchmark. *Advances in Neural Information Processing Systems* 37: 126500–126514.
- Vrettos M, Reynolds C, Amar A. 2021. Malar stripe size and prominence in peregrine falcons vary positively with solar radiation: Support for the solar glare hypothesis. *Biology Letters* 17: 20210116.
- Wenk E, Mesaglio T, Keith D, Cornwell W. 2024. Curating protected area-level species lists in an era of diverse and dynamic data sources. *Ecological Informatics* 84: 102921.
- Wolfe BW et al. 2025. Continental-scale assessment of climate-driven marine species range extensions using a decade of citizen science data. *Diversity and Distributions* 31: e70022.
- Wu SH, Chang HW, Lin RS, Tuanmu MN. 2022. SILIC: A cross database framework for automatically extracting robust biodiversity information from soundscape recordings based on object detection and a tiny training dataset. *Ecological Informatics* 68: 101534.
- Yoder JB, Andrade AK, DeFalco LA, Esque TC, Carlson CJ, Shryock DF, Yeager R, Smith CI. 2024. Reconstructing 120 years of climate change impacts on Joshua tree flowering. *Ecology Letters* 27: e14478.

Received: 30 March 2026. Revised: 4 August 2026. Accepted: 14 August 2026

© The Author(s) 2026. Published by Oxford University Press on behalf of the American Institute of Biological Sciences. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (<https://creativecommons.org/licenses/by-nc/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com